

Title	A data-driven approach to predict the success of bank telemarketing
Harvard reference	Moro, S., Cortez, P. and Rita, P., 2014. A data-driven approach to predict the success of bank telemarketing. <i>Decision Support Systems</i> , 62, pp.22-31.
Link	https://doi.org/10.1016/j.dss.2014.03.001 (Dataset available at: https://archive.ics.uci.edu/dataset/222/bank+marketing)
Modelling	The authors evaluated four distinct data mining methods: logistic regression, decision trees, support vector machines (SVM), and artificial neural networks (ANN). The neural network model achieved the highest predictive performance (measured via AUC) when identifying clients most likely to convert and subscribe to a term deposit.
Preprocessing techniques	The study utilized a semi-automatic feature selection process prior to modelling to reduce dimensionality, scaling down an initial pool of 150 client and socioeconomic attributes to a critical set of 22 predictive features. Temporal data and categorical attributes were transformed for algorithm compatibility.
Datasets used	The UCI Bank Marketing Dataset (Publicly Available). This dataset contains direct marketing campaign data from a Portuguese retail institution (2008 to 2013), featuring client demographics, temporal campaign data, macroeconomic indicators, and a binary conversion target variable.
Key takeaways	1. Rigorous feature selection is essential before modelling to isolate true behavioural signals from demographic noise. 2. This public dataset serves as an excellent benchmark to practice handling class imbalances and building baseline classification models before applying them to STADIOEquities project.

Table 2 Showing details of second publication

Title	Machine Learning Algorithms for Predicting Fintech Adoption Patterns: Insights from Blockchain-Based Payment Systems
Harvard reference	Hasibuan, A., et al., 2025. Machine Learning Algorithms for Predicting Fintech Adoption Patterns: Insights from Blockchain-Based Payment Systems. <i>ResearchGate</i> .
Link	https://www.researchgate.net/publication/397872897_Machine_Learning_Algorithms_for_Predicting_Fintech_Adoption_Patterns_Insights_from_Blockchain-Based_Payment_Systems

Modelling	The researchers implemented and compared Random Forest, Gradient Boosting Machines (GBM), and Neural Networks. The dataset was partitioned into a 70/30 train-test split. The ensemble methods (Random Forest and GBM) achieved an 87-89% predictive accuracy, significantly outperforming traditional regression models by capturing non-linear relationships.
Preprocessing techniques	Data collection involved aggregating 127 diverse variables across multiple domains (demographics, organizational factors, technological infrastructure, and behavioural intentions). Preprocessing involved cleaning survey data and transforming categorical perceptions into model-ready numerical formats.
Datasets used	A primary, proprietary dataset comprising records of 2,847 vocational workers across the manufacturing, retail, and service sectors in Indonesia, gathered between January 2024 and August 2024.
Key takeaways	1. Tree-based ensemble methods (like the XGBoost algorithm planned for STADIOEquities) are highly effective at capturing the non-linear behavioural relationships inherent in fintech user adoption. 2. Behavioural intentions and contextual friction points are far stronger predictors of user conversion than static demographic attributes alone.

Table 3 Containing information about third publications

Title	B2B Customer Conversion Prediction: A Document Representation, Graph Theory, and CatBoost Driven Methodology
Harvard reference	Wang, T., Azam, S.S., Huang, W.E., Wiranata, A., Brinton, C.G. and Allebach, J.P., 2026. B2B Customer Conversion Prediction: A Document Representation, Graph Theory, and CatBoost Driven Methodology. <i>arXiv preprint arXiv:2609.03239</i> .
Link	https://doi.org/10.48550/arXiv.2609.03239
Modelling	The researchers utilized a CatBoost classifier (a gradient boosting decision tree algorithm fundamentally like XGBoost) to predict customer conversion. The framework achieved a 91% prediction accuracy when identifying users most likely to successfully convert during complex, multi-step sales cycles.
Preprocessing techniques	The authors engineered a set of structured features by applying document representation and graph theory to map

	the sequential steps of the user's journey before feeding the data into the machine learning engine.
Datasets used	A proprietary dataset capturing long-term customer journey touchpoints, tracking users from initial interaction through to the final conversion outcome.
Key takeaways	1. Tree-based gradient boosting algorithms (like CatBoost or XGBoost) are highly robust and achieve strong accuracy when predicting binary conversion outcomes. 2. Transforming the specific, sequential steps of a user's journey (like tracking exactly where they drop off) into structured data features is a critical preprocessing step for model accuracy.

Table 4 Giving further details about the chosen dataset

Name	UCI Bank Marketing Dataset
Havard reference	Moro, S., Cortez, P. and Rita, P., 2014. A data-driven approach to predict the success of bank telemarketing. <i>Decision Support Systems</i> , 62, pp.22-31.
Link	https://archive.ics.uci.edu/dataset/222/bank+marketing
Number of instances	41,188 instances in the main split (bank-additional-full.csv), with a smaller 10% subset (bank-additional.csv) containing 4,119 instances for computationally heavy algorithms.
Number of feature columns	20 input feature columns (excluding the target variable).
Number of classes (if applicable)	2 classes (Binary classification): • no (Class 0): Client did not subscribe to a term deposit. • yes (Class 1): Client subscribed to a term deposit
Target range (if applicable)	Not Applicable (This is a classification problem, not regression).
Origins	Real-world historical data collected from direct marketing campaigns (phone calls) conducted by a Portuguese banking institution between May 2008 and November 2010.